

CH4207

Chemistry of Materials

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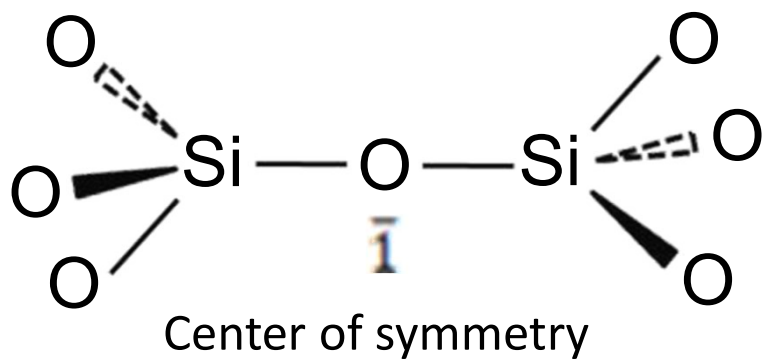
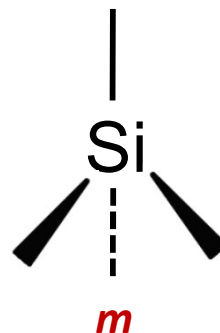
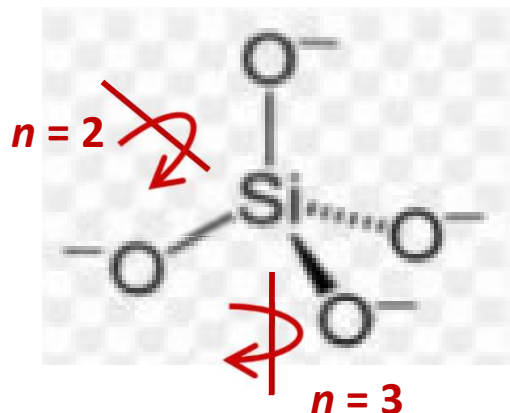
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SYMMETRY OF SOLIDS

(symmetry elements, point groups and space groups)

Symmetry elements

$[\text{SiO}_4]^{4-}$ is the conjugate base of silicic acid $\text{Si}(\text{OH})_4$



The axis about which the tetrahedron may be rotated is called **symmetry element** and the process of operation is called **symmetry operation**.

Symmetry elements

	Symmetry element	Hermann-Mauguin (Crystallography)	Schönflies (Spectroscopy)
Point symmetry	Mirror plane	m	σ_v, σ_h
	Rotation axis	n ($= 2, 3, 4, 6$)	C_n ($= C_2, C_3$ etc.)
	Inversion axis	$\bar{n}$ ($= \bar{1}, \bar{2}$, etc.)	---
	Alternating axis (rotoreflexion)	----	S_n ($= S_2, S_3$ etc.)
	Center of Symmetry	$\bar{1}$	i
Space symmetry	Glide plane	n, d, a, b, c	---
	Screw axis	$2_1, 3_1$, etc.	---

$n = 5, 7$ are never observed, but pentagonal symmetry do exist in nature (quasicrystals)
Pentagons cannot be packed to form a complete close packed lattice

Inversion axis, $\bar{n}$ is a combined symmetry operation involving rotation (according to n) and inversion through the center

Symmetry elements

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	Screw axis	$2_1, 3_1$, etc.	---

Space symmetry → Point symmetry elements + incremental translational steps

Screw axis → **X_Y** → **Rotation** by **$360^\circ/X$** + **translation** of the unit cell edge by **Y/X**

e.g. 4_2 axis is parallel to a involves translation by $a/2$ and rotation by 90°

Symmetry elements

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Space symmetry	Glide plane	n, d, a, b, c	---
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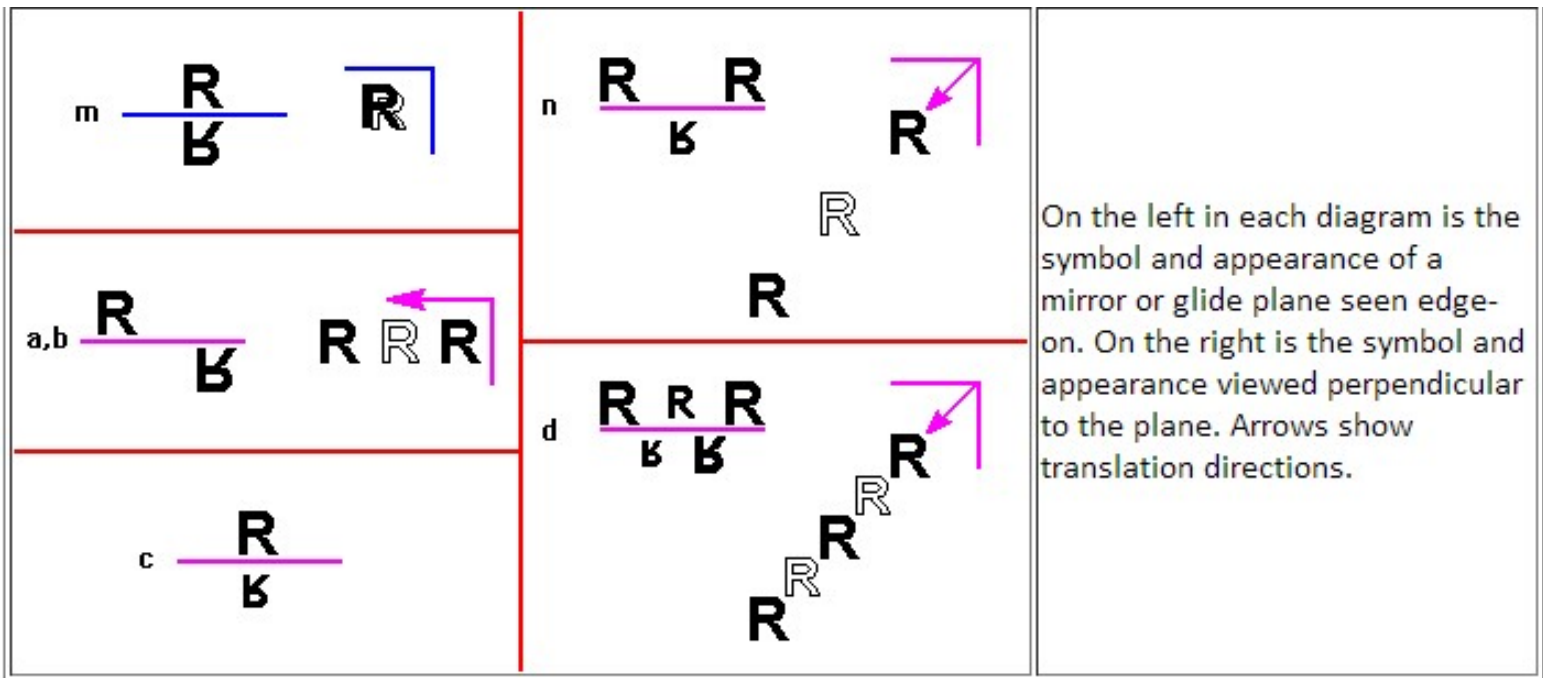
Space symmetry $\rightarrow$ Point symmetry elements + incremental translational steps

Glide plane $\rightarrow$ **Translation + reflection**

Translation may be parallel to any of the unit cell axes (a, b, c), to face diagonal (n), or to body diagonal (d)

a, b, c, n , glide planes all have translation step half the unit cell

Mirror and Glide Planes



On the left in each diagram is the symbol and appearance of a mirror or glide plane seen edge-on. On the right is the symbol and appearance viewed perpendicular to the plane. Arrows show translation directions.

m is a conventional mirror plane. Objects are reflected across the plane. Looking perpendicular to the plane we see the object and the reflection of its reverse side.

a, b are glides parallel to the unit cell edges. We see the object alternating with its translated reflection.

c is a glide parallel to the third edge of the unit cell. Since all the figures (except isometric classes) view down this direction, there is no view perpendicular to the plane. The object and its reflection are translated along the line of sight, so we see the object, then its reflection translated away from us (hence smaller). More distant translations are hidden behind (beneath) the two images shown.

n is a diagonal glide, half a unit cell edge in each direction. In the view along the plane, additional images would continue to step down and to the left, but they are hidden behind nearer images of the object.

d is like **n** in being a diagonal glide, but here the step is one quarter unit cell edge in each direction. Viewing along the plane we see four progressively more distant images of the object before the series in the neighboring unit cell begins.

32 Point Groups

Many molecules & crystals possess more than one element of point symmetry, but the number of combinations of symmetry elements that may occur in crystals is limited to 32 → **crystallographic point groups**

Crystal Class	Bravais Lattices	Point Groups
Triclinic	P	1, 1*
Monoclinic	P, C	2, m, 2/m
Orthorhombic	P, C, F, I	222, mm2, 2/m 2/m 2/m
Trigonal	P, R	3, 3*, 32, 3m, 3*2/m
Hexagonal	P	6, 6*, 6/m, 622, 6mm, 6*m2, 6/m 2/m 2/m
Tetragonal	P, I	4, 4*, 4/m, 422, 4mm, 4*2m, 4/m 2/m 2/m
Isometric	P, F, I	23, 2/m3* 432, 4*3m, 4/m 3* 2/m

* means bar

Taken from: <https://www.uwgb.edu/dutchs/SYMMETRY/3dSpaceGrps/3dspgrp.htm>

Point Symmetry Elements

Symmetry elements	Symbol (written)	Graphical symbol
Rotation axes	1	None
	2	
	3	
	4	
	6	
Mirror plane	m	

Symmetry elements	Symbol (written)	Graphical symbol
Inversion axes	$\bar{1}$	
	$\bar{2} (\equiv m)$	
	$\bar{3} (\equiv 3 + \bar{1})$	
	$\bar{4}$	
	$\bar{6} (\equiv 3/m)$	

Point Symmetry Elements

None	1
	2
	2 ₁
	3
	3 ₁
	3 ₂
	4
	4 ₁
	4 ₂
	4 ₃
	6
	6 ₁
	6 ₂
	6 ₃
	6 ₄
	6 ₅

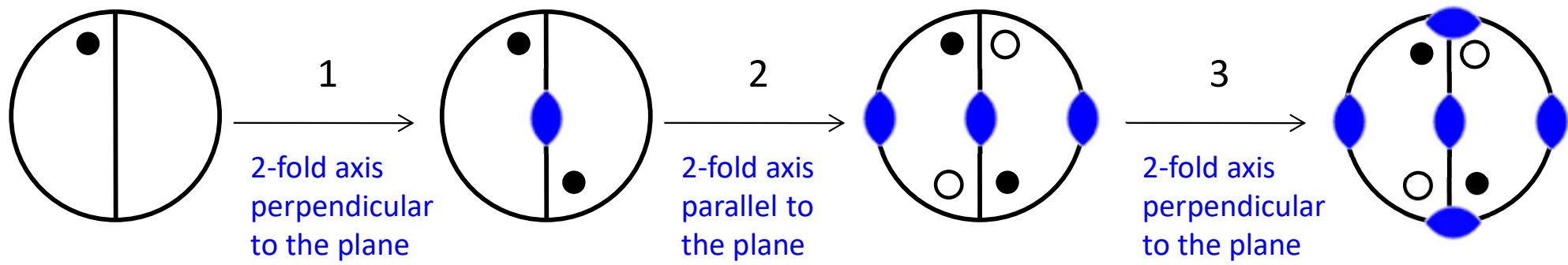
	$\bar{1}$
	2/m
	2 ₁ /m
	$\bar{3}$
	$\bar{4}$
	4/m
	4 ₂ /m
	$\bar{6}$
	6/m
	6 ₃ /m
	m
	a,b,c
	a,b,c
	n
	$\frac{1}{8}$ d
	$\frac{3}{8}$ d

Rotation axes are represented with conventional crystallographic symbols.

Rotation axes parallel to the plane of the diagram are indicated by a symbol and a line showing the axis orientation. Apart from isometric space groups, only 2-fold axes ever have this orientation.

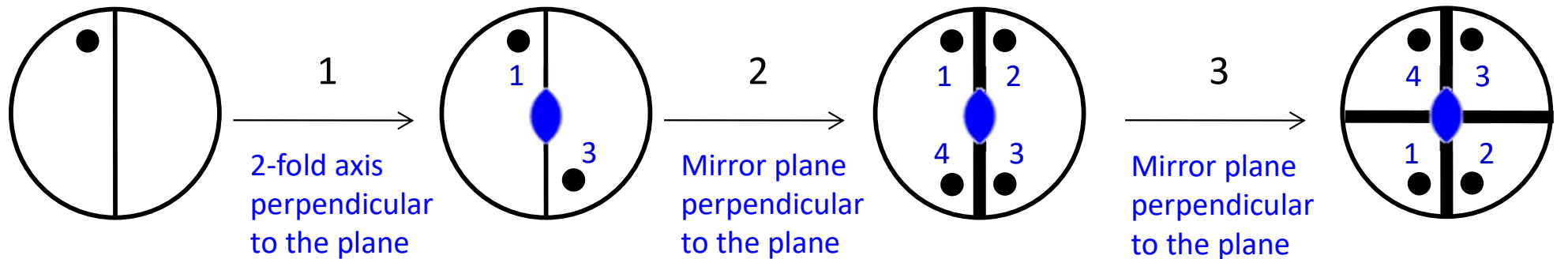
Point Symmetry: Orthorhombic

222

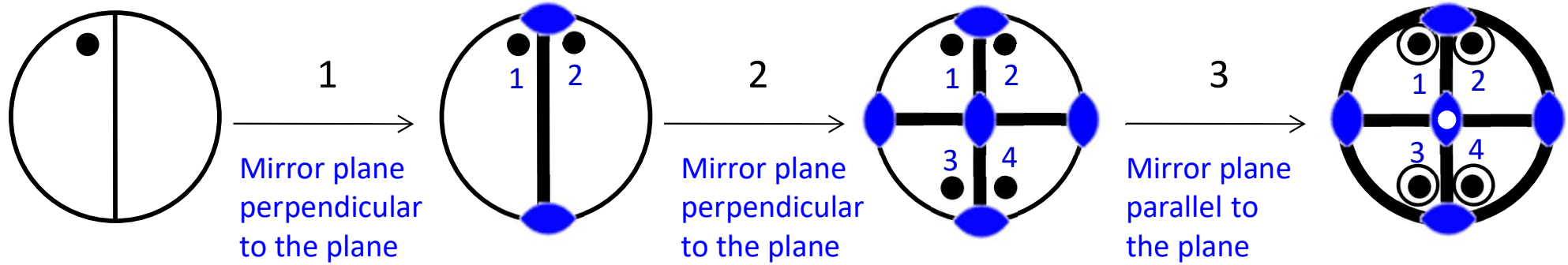


3rd step is dependent; we can also write **22**

mm2



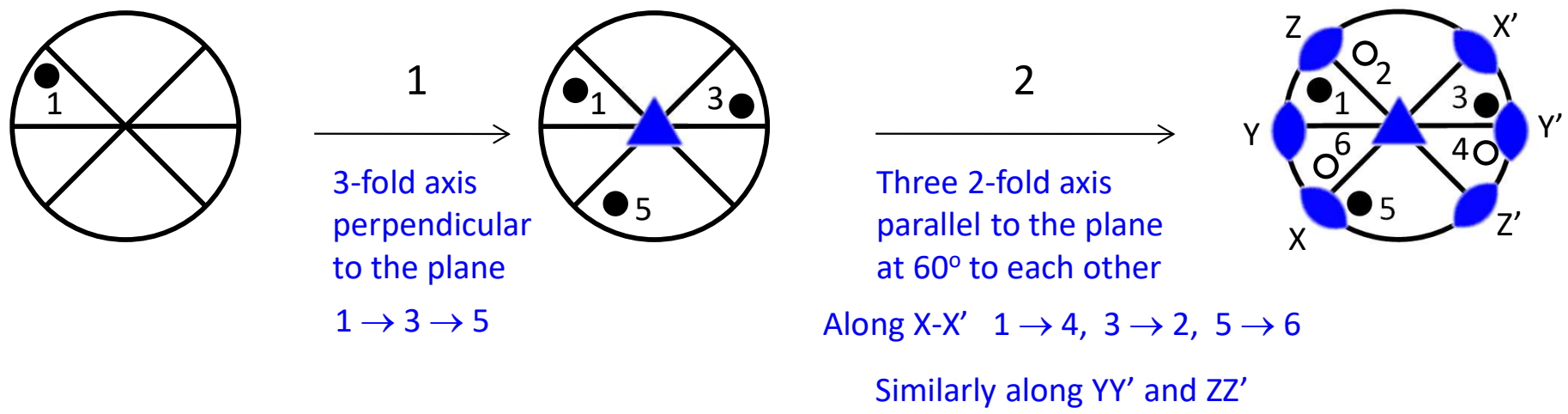
mmm $\equiv$ **2/m 2/m 2/m**



centrosymmetric

Point Symmetry: Trigonal

32



$4/mmm \rightarrow (4/m)mm \rightarrow$ 4-fold axis has a mirror plane perpendicular to it

Space group

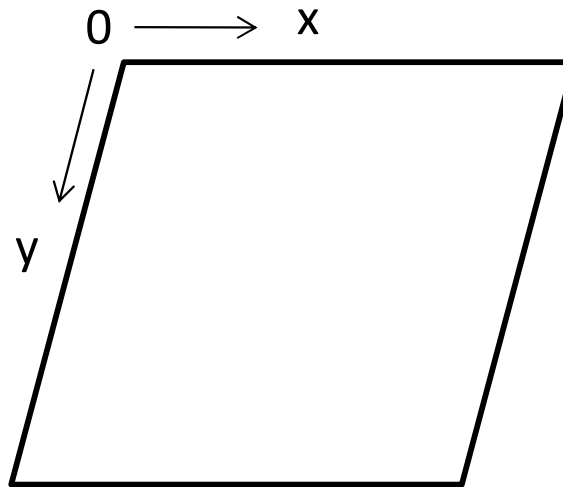
32 crystallographic point groups:

27 non-cubic

73 symmorphic (No. of space groups without using glide plane / screw axis)

Convention to label axes in space group

Space group is drawn as parallelogram with the plane corresponding to xy plane of the unit cell
 z -axis points out of the plane



Space group

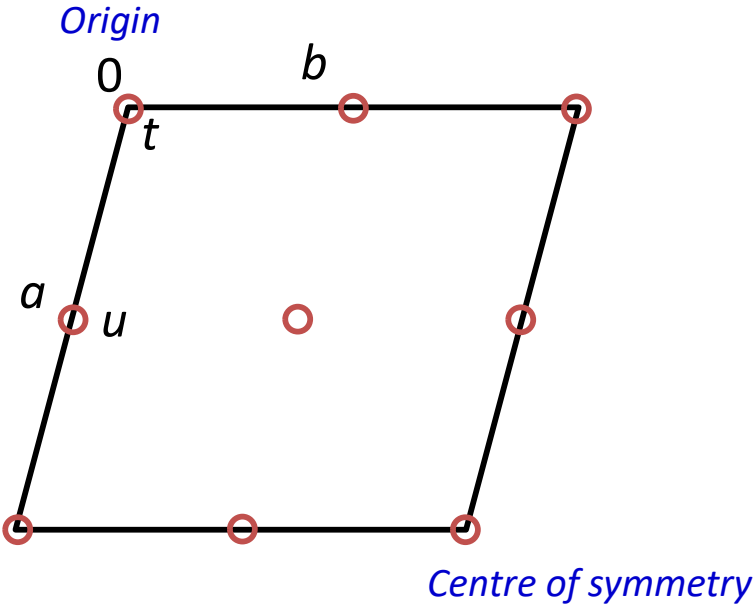
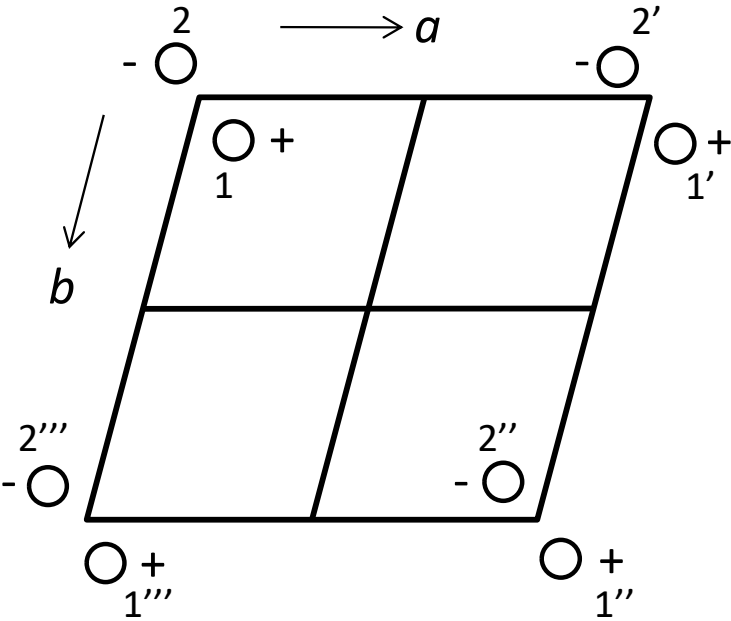
For each space group, two parallelograms are used:
Left: give equivalent positions & Right: give symmetry elements

Triclinic

$P\bar{1}$

+ : positive z axis
- : negative z axis

The starting position 1 is present in other unit cells as 1', 1'', 1'''



Space group

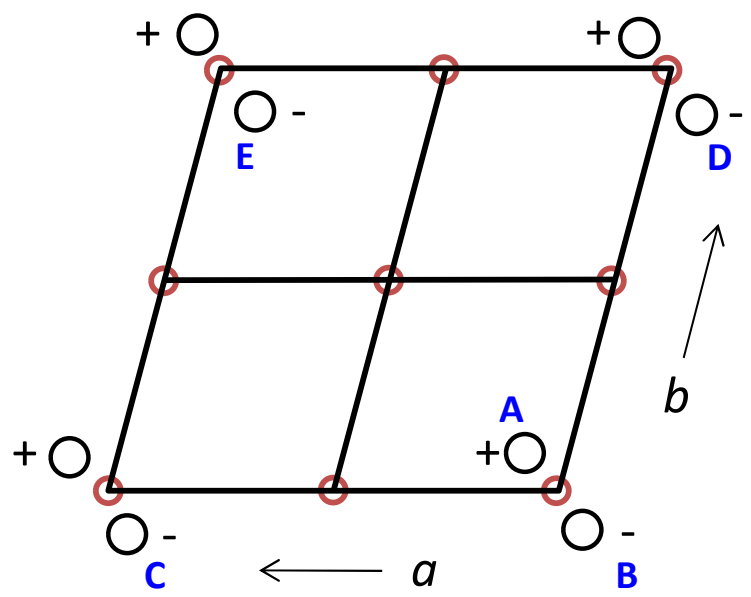
Triclinic

$P\bar{1}$

+ : positive z axis
- : negative z axis

Projection down the c-axis

The unit cell angles are not orthogonal by symmetry



Space group **$P1$** :
only one symmetry equivalent position within the unit cell $\rightarrow x, y, z$

Space group **$P\bar{1}$** :
two symmetry equivalent positions due to the presence of the point of inversion $\rightarrow x, y, z$ and $\bar{x}, \bar{y}, \bar{z}$

This does not imply a single point of inversion per unit cell

A (x, y, z coordinates) is symmetry related to **B** ($\bar{x}, \bar{y}, \bar{z}$) by the point of inversion at the origin

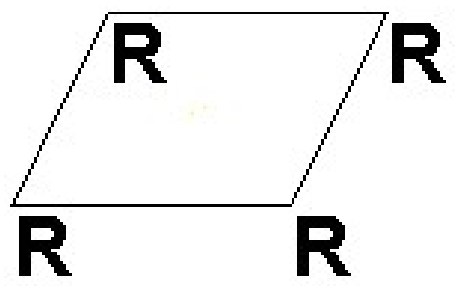
- | | |
|--|---|
| Lattice translation of B along $\vec{a}$ brings it to C ($1-x, -y, -z$) | Point of inversion: $\frac{1}{2}, 0, 0$ |
| Lattice translation of B along $\vec{b}$ brings it to D ($-x, 1-y, -z$) | Point of inversion: $0, \frac{1}{2}, 0$ |
| Lattice translation of B along $\vec{a}$ and $\vec{b}$ brings it to E ($1-x, 1-y, -z$) | Point of inversion: $\frac{1}{2}, \frac{1}{2}, 0$ |

Other points of inversion (e.g. the one at $1, 0, 0$) lie in neighboring unit cells
Translation along $\vec{c}$: 4 more points of inversion. There are total 8 points of inversion per primitive unit cell

Fractional Coordinates	0,0,0	1/2,0,0	0,1/2,0	1/2,1/2,0	0,0,1/2	1/2,0,1/2	0,1/2,1/2	1/2,1/2,1/2
Symmetry Operator	-x,-y,-z	1-x,-y,-z	-x,1-y,-z	1-x,1-y,-z	-x,-y,1-z	1-x,-y,1-z	-x,1-y,1-z	1-x,1-y,1-z

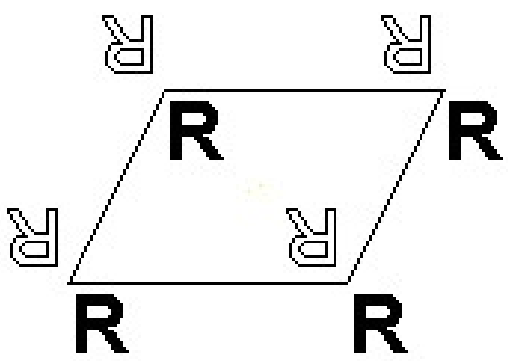
Triclinic

P1



1. P1
 (x,y,z)

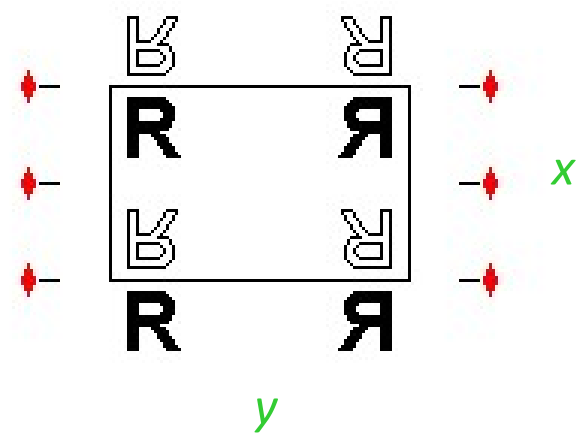
P $\bar{1}$



2. P1'
 $(x,y,z); (-x,-y,-z)$

Monoclinic (2)

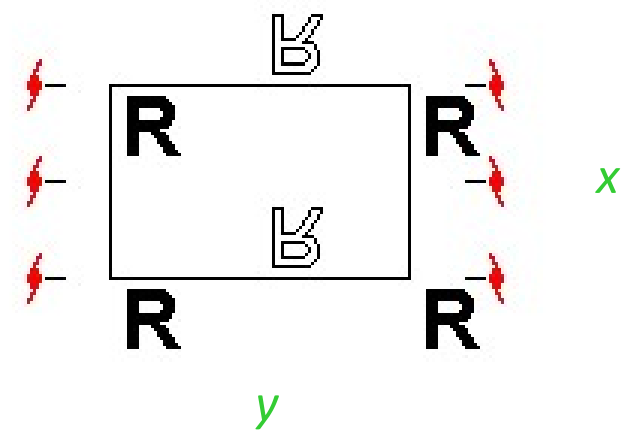
P2



P2

$(x,y,z); (-x,y,-z)$

P2₁

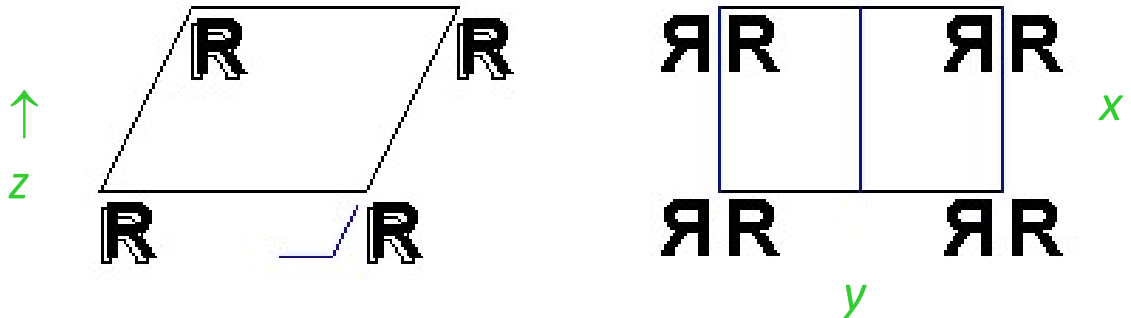


P2₁

$(x,y,z); (-x,1/2+y,-z)$

Monoclinic (m)

Pm



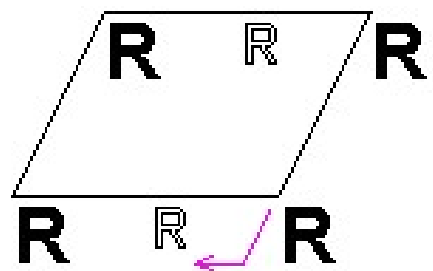
6. Pm

$(x,y,z); (x,y,-z)$

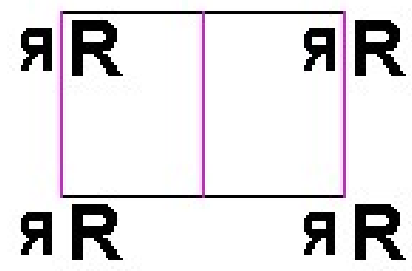
Pm

$(x,y,z); (x,-y,z)$

Pb



Pc



7. Pb

$(x,y,z); (x,1/2+y,-z)$

Pc

$(x,y,z); (x,-y,1/2+z)$